

Part 01 : Theoretical Questions

Q1 : What is the difference between static binding and dynamic binding? When does each one happen?

Q2 : What is the difference between method overloading and method overriding?

Q3 : What keywords are used for Method Overriding? What does each one mean ?

Part 02 : Practical (Extending the Movie Ticket Booking System)

In the previous assignments, you built a Movie Ticket Booking System with inheritance, properties, and static members. Now you will apply polymorphism to make the system flexible and extensible.

What you need to build :

1. Refactor the base Ticket class:

- a. Add a PrintTicket() method that prints: TicketId, MovieName, Price, PriceAfterTax. Child classes should be able to provide their own version of this method.
- b. Add two versions of a SetPrice method — one that takes a decimal (sets price directly) and one that takes a decimal base price and a decimal multiplier (sets price = base × multiplier).

2. In each child class, provide its own version of PrintTicket():

- a. StandardTicket — prints the base ticket info and the SeatNumber.
- b. VIPTicket — prints the base ticket info, LoungeAccess, and ServiceFee.
- c. IMAXTicket — prints the base ticket info and whether it is 3D.

3. In the Cinema class, update PrintAllTickets() so it loops through the Ticket[] array and calls PrintTicket() on each one.

4. Create a static method ProcessTicket(Ticket t) that takes any Ticket and calls PrintTicket() on it.

5. In Main:

- a. Create a Cinema and open it.
- b. Create one StandardTicket, one VIPTicket, and one IMAXTicket.
- c. Test both versions of SetPrice on one ticket.
- d. Add all tickets to the Cinema and call PrintAllTickets().
- e. Call ProcessTicket() with one of the tickets.
- f. Close the Cinema.

Expected Output (Example) :

```
===== Cinema Opened =====
Projector started.

===== SetPrice Test =====
Setting price directly: 150
Setting price with multiplier: 100 x 1.5 = 150

===== All Tickets =====
Ticket #1 | Inception | Price: 150 EGP | After Tax: 171.00 EGP
    Seat: A-5
Ticket #2 | Avengers | Price: 200 EGP | After Tax: 228.00 EGP
    Lounge: Yes | Service Fee: 50 EGP
Ticket #3 | Dune | Price: 180 EGP | After Tax: 205.20 EGP
    IMAX 3D: No

===== Process Single Ticket =====
Ticket #2 | Avengers | Price: 200 EGP | After Tax: 228.00 EGP
    Lounge: Yes | Service Fee: 50 EGP

===== Cinema Closed =====
Projector stopped.
```